

Technical Report: Structural Analysis and Design Optimization of a Single-Cell Wingbox

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1. Abstract

This report details the structural analysis and optimization of a single-cell wingbox subjected to aerodynamic and inertial loading. The study compares the performance of two materials—**Aluminum 7075-T6** and **Carbon/Epoxy composite**—across varying flight conditions, including Cruise, Positive High Angle of Attack (PHAA), Positive Low Angle of Attack (PLAA), Negative High Angle of Attack (NHAA), and Negative Low Angle of Attack (NLAA). Using a custom MATLAB analysis tool, we evaluated cross-sectional properties, shear flow distribution, bending stresses, and margin of safety (MS). The final design analysis demonstrates that the Carbon/Epoxy configuration offers superior specific strength, achieving a weight reduction of approximately **88.4%** compared to the Aluminum counterpart while maintaining positive margins of safety.

2. Methodology

2.1. Analytical Approach

The structural analysis was performed using a discretized thin-walled beam model. The wing cross-section was idealized as a single-cell box reinforced by four stringers. The analysis followed a four-step iterative design process:

1. **Preliminary Analysis (Carbon/Epoxy):** Initial sizing and stress check.
2. **Design Analysis (Aluminum):** Sizing required to meet safety factors using isotropic materials.
3. **Structural Optimization (Carbon/Epoxy):** Minimizing weight by reducing stringer area and skin thickness while maintaining $MS > 0$.
4. **Final Design Comparison:** Quantitative comparison of weight and stiffness.

2.2. Governing Equations

The analysis utilized **Thin-Walled Beam Theory** and **Modulus-Weighted Section Properties** to account for composite material behavior.

- **Modulus Weighted Centroid:**

$$\bar{Y} = \frac{\sum E_i A_i y_i}{\sum E_i A_i}, \quad \bar{Z} = \frac{\sum E_i A_i z_i}{\sum E_i A_i}$$

- **Bending Stress (Axial):**

Axial stress (σ_{xx}) in the stringers was calculated using the combined bending moment equations:

$$\sigma_{xx} = E \left[\frac{P_x}{EA} - \frac{M_z I_{yy} + M_y I_{yz}}{I_{yy} I_{zz} - I_{yz}^2} y + \frac{M_y I_{zz} + M_z I_{yz}}{I_{yy} I_{zz} - I_{yz}^2} z \right]$$

- **Shear Flow (\$q\$):**

Shear flow distribution was determined by solving the system of linear equations derived from the jump matrix for a multi-cell (or single-cell) structure:

$$q(s) = q_0 - \frac{V_y}{I_{xx}} Q_z(s) - \frac{V_z}{I_{xx}} Q_y(s)$$

$$q_{matrix} \cdot q = \text{Force Vector}$$

- **Margin of Safety (MS):**

$$MS = \frac{\sigma_{allowable}}{\sigma_{applied}} - 1$$

3. Material Properties

Property	Aluminum 7075-T6	Carbon/Epoxy	Units
Density	0.10	0.056	\$lb/in^3\$
Young's Modulus	10	23.5	Msi
Shear Modulus	3.75	6	Msi
Yield Strength - Tension	37	263	Ksi
Ultimate Strength - Tension	43	263	Ksi
Yield Strength - Compression	-37	-263	Ksi
Ultimate Strength - Compression	-43	-263	Ksi
Yield Strength - Shear	24	88	Ksi
Ultimate Strength - Shear	28	88	Ksi

4. Results and Discussion

4.1. Preliminary vs. Design Analysis (Carbon/Epoxy)

Initial analysis of the Carbon/Epoxy wing (Step 1) yielded a minimum wing weight of **154.65 lb** with high margins of safety, specifically a minimum stringer MS of **0.2838** in the PLAA condition.

Subsequent optimization (Step 3) reduced the stringer area to **1.19 in²** and skin thickness to **0.018 inches**. This optimized configuration resulted in a final weight of **92.36 lb**. Despite the mass reduction, the structure maintained a positive margin of safety $MS \approx 0.0015$ under the critical PLAA loading condition.

4.2. Design Analysis (Aluminum 7075-T6)

The Aluminum wing design required significantly more material to sustain the same loads.

- **Required Stringer Area:** 6.5376 in².
- **Required Skin Thickness:** 0.05534 inches.
- **Total Weight:** 797.29 lb.

The Aluminum design suffered from lower specific stiffness and strength. To maintain a positive $MS \approx 1.5 \times 10^{-5}$ for PLAA stringers, the aluminum structure required **~5.5x** the stringer cross-sectional area and **~3x** the skin thickness compared to the composite design.

4.3. Displacement and Twist

- **Carbon/Epoxy (Optimized):** Maximum tip vertical displacement was **11.19 inches** in the PHAA/PLAA conditions.
- **Aluminum:** Maximum tip vertical displacement was **3.58 inches**.
- **Analysis:** While the Aluminum wing is stiffer (less displacement) due to the massive increase in material volume required for strength, it comes at an unacceptable weight penalty. The Carbon wing is more flexible but remains within structural limits.

5. Conclusion

The comparative analysis highlights the critical advantage of advanced composites in aerospace structures.

- **Weight Savings:** The optimized Carbon/Epoxy wing weighs **92.36 lb**, whereas the viable Aluminum wing weighs **797.29 lb**. This represents a weight factor difference of approximately **8.6x**.
- **Efficiency:** The Carbon/Epoxy material allows for a stringer area **1/9th** the size and a skin thickness **1/5th** the size of the Aluminum equivalent while maintaining structural integrity.

The Carbon/Epoxy design is validated as the superior choice for this wingbox application, offering a high strength-to-weight ratio essential for aircraft performance and fuel efficiency.